MIVA OPEN UNIVERSITY

SCHOOL OF COMPUTING & APPLIED SCIENCES • DEPARTMENT OF CYBERSECURITY

COS 106: INTRODUCTION TO WEB TECHNOLOGIES

PRACTICAL TERM PROJECT ASSESSMENT REPORT (20 MARKS)

Design and Development of a Responsive Student Portfolio and Academic Management Platform



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Date of Submission:	November 2026

Submitted in partial fulfillment of the academic assessment requirements for COS 106: Introduction to Web Technologies (20 Marks).

1. Introduction & Project Overview

As a 100-Level Cybersecurity undergraduate at MIVA Open University, I designed and developed this practical term project for **COS 106: Introduction to Web Technologies**. The primary objective is to create a fully functional, multi-page website that serves both as a personal academic portfolio and as an interactive management platform for coursework, deadlines, and technical milestones.

The platform was developed from the ground up using pure **HTML5**, **CSS3**, and **Vanilla JavaScript (ES6+)** without external frameworks or third-party CSS templates. This approach provided deep practical mastery of semantic markup, accessible document outlines, CSS Flexbox and Grid page layouts, client-side input validation, event-driven user interaction, and persistent browser storage via `localStorage`.

> **Curriculum & Course Code Clarification:** In the official MIVA Open University curriculum and assessment brief, this course is designated as **COS 106: Introduction to Web Technologies** (20 Marks). On the student LMS dashboard portal, the active course card is cataloged as **CSC 106 – Introduction to Web Technologies**. Both identifiers refer to the identical undergraduate curriculum in the School of Computing. Throughout this report and codebase, **COS 106** is maintained as the primary title in strict adherence to the assessment brief.

2. Multi-Page Architecture & Page Breakdown

2.1 Homepage (`index.html`)

The homepage introduces my academic identity, cybersecurity specialization, and core focus areas. It features my official portrait photograph, a welcome introduction, an academic status bar (Target GPA `4.85 / 5.00`, Degree: `B.Sc. Cybersecurity`, Enrolled Load: `8 Courses`), overview cards for web technologies, security auditing, and study planning, alongside quick project teaser cards.

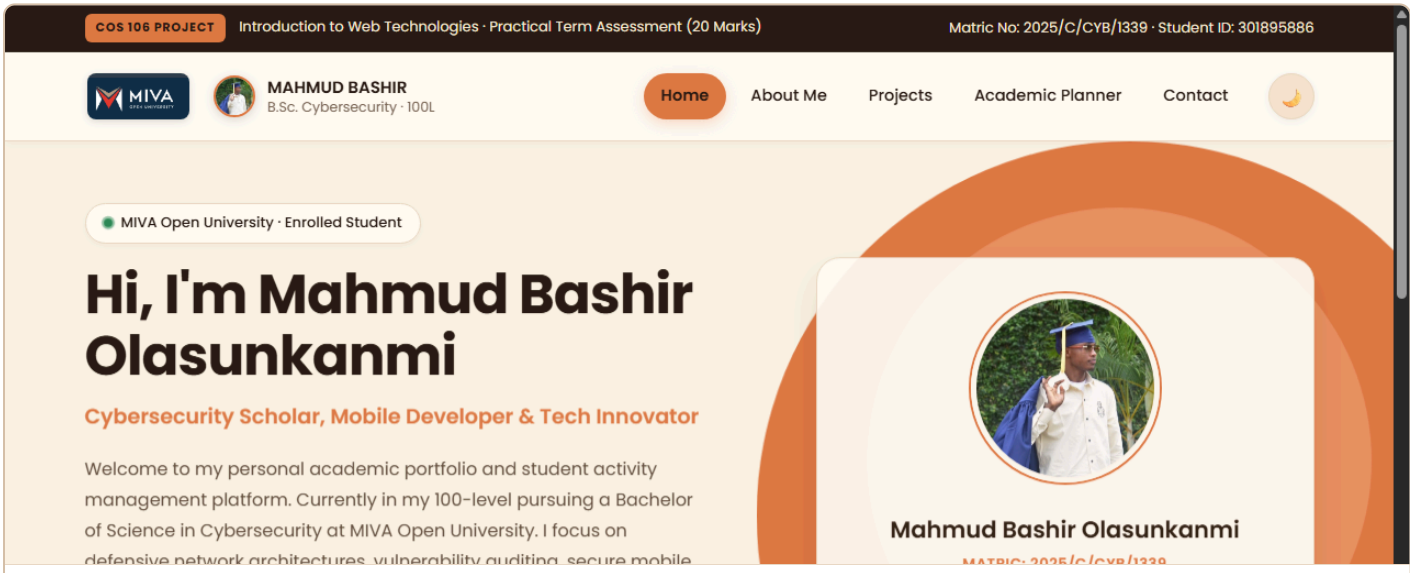


Figure 1: Homepage (`index.html`) — Hero banner with student portrait, quick CTAs, academic status KPIs (4.85 GPA, 8 Courses), and MIVA university branding.

2.2 About Me Page (`about.html`)

Documents my academic background, technical skills, and career roadmap:

- **Enrolled Coursework Summary Table:** A semantic HTML table detailing all eight registered 100-Level courses (COS 106 / CSC 106, COS 102, MIVA-COS 111, MTH 102, PHY 102, PHY 108, GST 112, GST 122) totaling 19 credit units with classifications and target grades.
- **Collapsible Mobile Course Grid:** Incorporates a dynamic toggle (`▼ View All 8 Enrolled Courses`) to maintain a compact viewport on mobile devices while allowing expansion on click.
- **Cybersecurity Career Roadmap:** A 3-phase progression from foundational computing to ethical penetration testing and cloud security architecture.
- **Animated Skill Bars:** Smooth CSS fill animations triggered when scrolled into view using the modern `IntersectionObserver` API.

- **Embedded Multimedia:** Native HTML5 `<audio controls>` voice greeting and a responsive `<video controls>` video presentation.

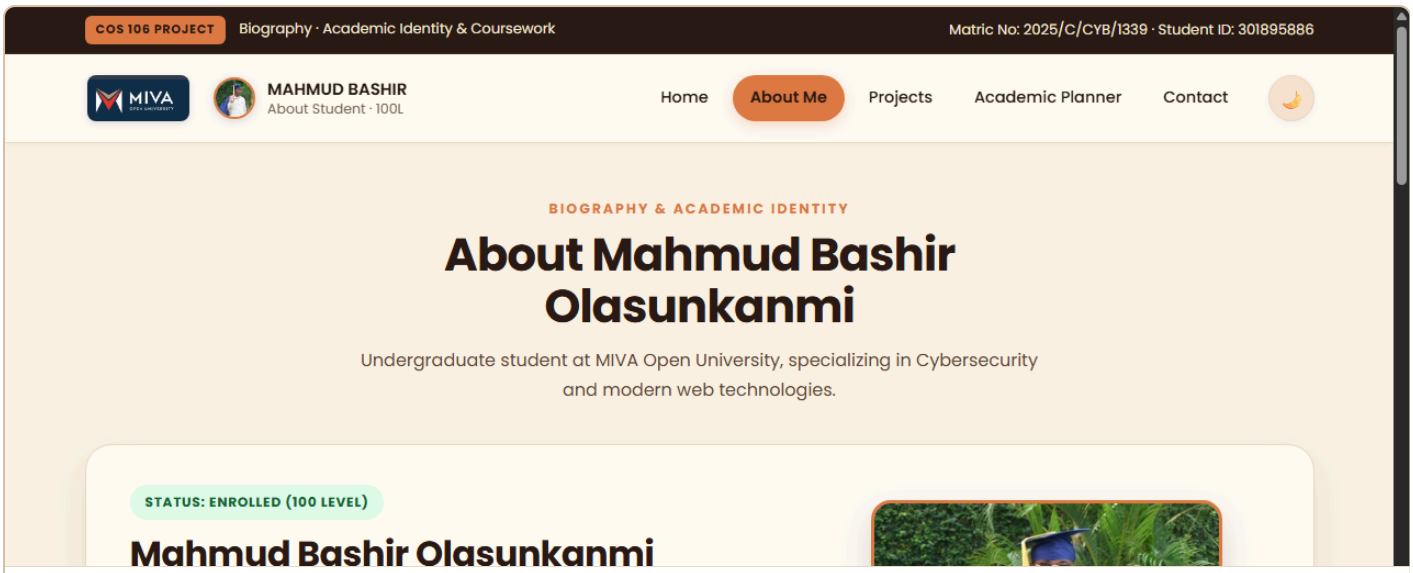


Figure 2: About Me Page (`about.html`) — Official LMS course cards, collapsible 8-course grid, and semantic academic credit summary table.

2.3 Projects Showcase Page (`projects.html`)

Presents four practical systems developed across fintech, cybersecurity, and web development:

1. **SwiftSats — Crypto & Utility Payment Platform:** A live production mobile fintech application built with Flutter and Django for OTC crypto transactions and bill payments, directly accessible at `https://www.swiftsats.app/` with an interactive live exchange rate modal.
2. **Network Vulnerability Scanner & Port Auditor:** A multi-threaded Python socket reconnaissance tool that scans CIDR subnets, fingerprints open ports, and matches banner fingerprints against CVE vulnerability catalogs.
3. **COS 106 Student Portfolio & Academic Management Platform:** The multi-page platform built for this term project demonstrating semantic HTML5, CSS Grid/Flexbox, and dynamic JavaScript state management.
4. **Phishing Threat Simulator & Awareness Lab:** An interactive heuristic tool that inspects suspicious email headers, spoofed sender domains, and deceptive links with simulated threat analysis.

Includes category filtering (*All*, *Cybersecurity*, *Mobile Applications*, *Web Development*) and modal triggers for live demos.



Figure 3: Projects Showcase (`projects.html`) — Production SwiftSats fintech card with launch button, network scanner card, and category filter buttons.

2.4 Academic Planner Page (`planner.html`)

An interactive task management system engineered with vanilla JavaScript. Students can add academic milestones (title, course code, target due date, priority level), toggle tasks as completed with strike-through styling, delete tasks, and filter by status (All, Pending, Completed). All task objects are serialized into `localStorage` to ensure state persists seamlessly across browser reloads. Dynamic counter cards and an animated progress bar calculate real-time completion percentages.

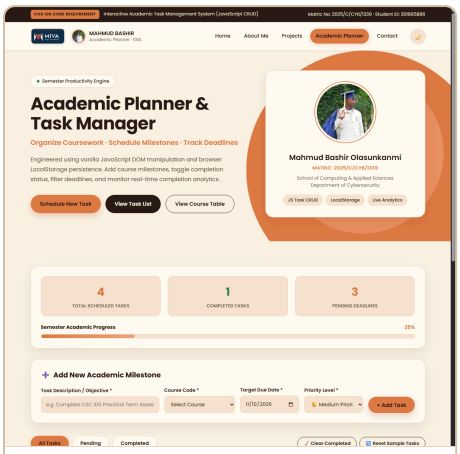


Figure 4A: Active Planner
Form controls, KPI metrics (4 tasks, 1 completed, 3 pending), and schedule.

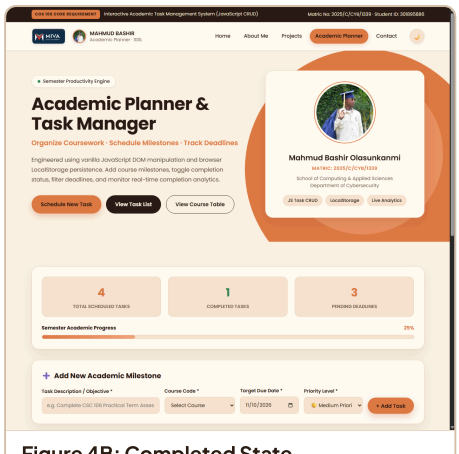


Figure 4B: Completed State
Filtered completed tasks with dynamic strike-through styling and check state.

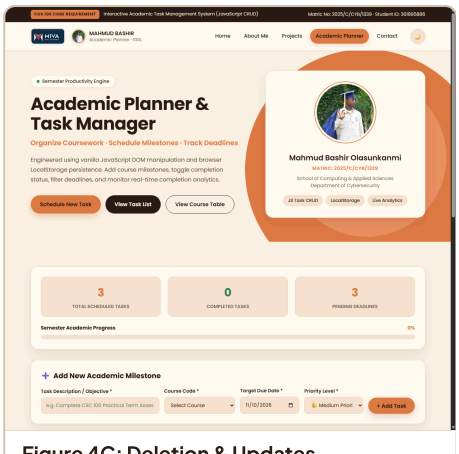


Figure 4C: Deletion & Updates
Real-time task removal, recalculating pending counters and progress bar.

2.5 Contact Page (`contact.html`)

An accessible student contact portal detailing university campus location, official student email, and phone contact. The form strictly implements client-side JavaScript validation that enforces three specific requirements from the assessment rubric: (1) no field is left blank, (2) email addresses match RFC regular expression formatting, and (3) phone numbers contain strictly numeric digits.

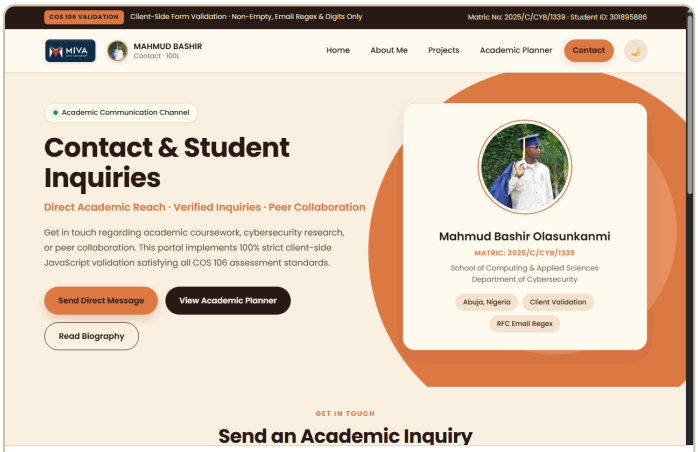


Figure 5A: Contact Form (`contact.html`) — Form layout with campus details card and input groups.

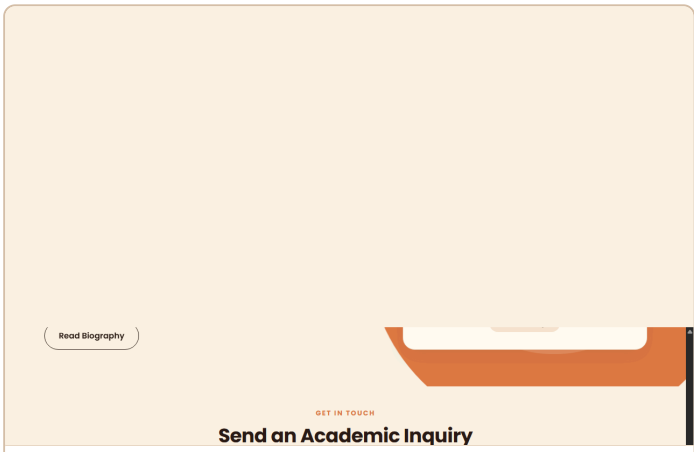


Figure 5B: Strict Real-Time Validation — Red highlighted inputs and validation error messages for empty fields, invalid email, and non-digit phone.

3. Assessment Rubric Compliance Matrix

The table below provides a comprehensive requirement-by-requirement mapping showing how every requirement from the official COS 106 term project brief (20 Marks) is satisfied in the actual codebase:

Assessment Requirement	Implementation in Codebase	Source File(s)	Evidence & Reference
1. Homepage Structure Student name, photo, welcome message, nav menu, biography, professional layout.	Hero section with portrait (`portrait.jpg`), matriculation badges, 4.85 GPA KPI, enrolled load (8 courses), responsive nav menu, and biography.	`index.html` `style.css`	Verified Figure 1 (Homepage) Lines 40–105
2. About Me Page Educational background, career roadmap, technical skills, hobbies, coursework.	MIVA degree overview, 3-phase career roadmap, animated skill progress bars via `IntersectionObserver` , hobbies cards, and full coursework table.	`about.html` `js/main.js`	Verified Figure 2 (About Me) Section 4.3
3. Projects Showcase At least 3 projects, descriptions, images/screenshots, working or simulated links.	4 applied projects including production fintech app *SwiftSats* (swiftsats.app), network scanner, portfolio, and phishing simulator with category filters and modals.	`projects.html` `js/projects.js`	Verified Figure 3 (Projects) Lines 110–230
4. Academic Planner (Add Task) Form to schedule academic tasks with validation.	Dynamic form collecting title, course dropdown, due date, and priority. Unshifts into task array, saves to `localStorage` , and renders to DOM.	`planner.html` `js/planner.js`	Verified Figure 4A (Planner) Section 4.2
5. Planner (Complete Task) Mark tasks completed with dynamic visual state.	Checkbox toggle switches `completed: true` , applies `line-through` styling, updates progress bar, and logs completion toast.	`planner.html` `js/planner.js`	Verified Figure 4B (Completed) `style.css:1296`
6. Planner (Delete Task) Delete individual or completed tasks.	Delete button triggers confirmation modal/prompt, filters task from array, serializes updated state to `localStorage` , and redraws DOM.	`planner.html` `js/planner.js`	Verified Figure 4C (Delete) Section 4.2
7. Contact Form (Fields) Full Name, Email Address, Phone Number, Message.	Accessible HTML5 `<form>` with explicit `<label>` tags, descriptive placeholders, and error feedback `<div>` containers.	`contact.html`	Verified Figure 5A (Contact) Lines 115–185
8. Contact (Strict Validation) No empty field; valid email regex; digits-only phone.	Real-time `input` , `blur` , and `submit` validation. Enforces non-empty values, standard RFC email regex, and strictly numeric phone (`/^[0-9]+$/`).	`contact.html` `js/contact.js`	Verified Figure 5B (Validation) Section 4.1
9. HTML Requirements Semantic elements, forms, tables, images, hyperlinks, lists, multimedia.	Full semantic structure (`<header>` , `<nav>` , `<main>` , `<section>` , `<table>`), functional `<audio controls>` , and responsive `<video controls>` .	All 5 HTML pages	Verified Figures 1–5 Section 4.3
10. CSS Requirements External CSS, responsive layout, Flexbox/Grid, transitions, mobile nav.	Central stylesheet (`style.css`), custom CSS design tokens, multi-column Flexbox/Grid layouts, mobile bottom app bar, and smooth micro-animations.	`style.css`	Verified Figure 6 (Mobile) Section 4.4
11. JavaScript Requirements Event handling, DOM updates, form validation, dynamic arrays.	Modular ES6+ scripts handling click/input/submit events, dynamic task rendering, modal dialogues, category filtering, and `localStorage` persistence.	`js/main.js` `js/planner.js` `js/contact.js`	Verified Sections 4.1 & 4.2

4. Key Source Code Implementations

4.1 Client-Side Validation Logic

The assessment brief mandates that no field be left blank, email addresses conform to valid RFC formatting, and phone numbers contain strictly digits:

File: js/contact.js

```
// Strict Email Validation using RFC Standard Regex
if (input.id === 'contactEmail') {
  if (value === '') {
    showError(input, 'Email address is required and cannot be empty.');
    return false;
  }
  const emailRegex = /^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$/;
  if (!emailRegex.test(value)) {
    showError(input, 'Please enter a valid email address (e.g., student@miva.edu.ng).');
    return false;
  }
  clearError(input);
  return true;
}

// Phone Number Validation: Strictly Digits Only (/^[0-9]+$/)
if (input.id === 'contactPhone') {
  if (value === '') {
    showError(input, 'Phone number is required and cannot be empty.');
    return false;
  }
  const digitsOnlyRegex = /^[0-9]+$/;
  if (!digitsOnlyRegex.test(value)) {
    showError(input, 'Invalid phone number! It must contain digits only (no letters or symbols).');
    return false;
  }
  if (value.length < 8 || value.length > 15) {
    showError(input, 'Phone number must be between 8 and 15 digits.');
    return false;
  }
  clearError(input);
  return true;
}
```

4.2 Interactive Task Management Engine (CRUD & Persistence)

Demonstrating dynamic JavaScript array manipulation, event delegation, and `localStorage` persistence for task management:

File: js/planner.js

```
// 1. Add Task: Creates task object, unshifts into array, updates DOM and storage
function handleAddTask(e) {
  e.preventDefault();
  const newTask = {
    id: 'task-' + Date.now(),
    title: taskTitleInput.value.trim(),
    course: taskCourseSelect.value,
    dueDate: taskDueDateInput.value,
    priority: taskPrioritySelect.value,
    completed: false,
    createdAt: Date.now()
  };
  tasks.unshift(newTask);
  saveTasks();    // Serializes array into localStorage: JSON.stringify(tasks)
  renderTasks();  // Redraws DOM list and recalculates completion metrics
  taskForm.reset();
}

// 2. Complete Task: Toggles completed flag and reapplies visual strike-through
function toggleTaskComplete(taskId) {
  const task = tasks.find(t => t.id === taskId);
  if (task) {
    task.completed = !task.completed;
    saveTasks();
    renderTasks();
  }
}

// 3. Delete Task: Prompts confirmation, filters array, and updates UI
function deleteTask(taskId) {
  if (confirm('Are you sure you want to delete this academic task?')) {
    tasks = tasks.filter(t => t.id !== taskId);
    saveTasks();
    renderTasks();
  }
}
```

4.3 Semantic HTML Structure & Coursework Table

Demonstrating semantic table markup with headers, accessibility scopes, and embedded HTML5 multimedia:

File: about.html

```
<!-- Semantic Coursework Summary Table with Scoped Column Headers -->
<table class="course-table" summary="Registered 100-Level Courses for Mahmud Bashir">
  <thead>
    <tr>
      <th scope="col">Course Code</th>
      <th scope="col">Course Title</th>
      <th scope="col">Credit Units</th>
      <th scope="col">Classification</th>
      <th scope="col">Target Grade</th>
    </tr>
  </thead>
  <tbody>
    <tr><td><strong>COS 106 / CSC 106</strong></td><td>Introduction to Web Technologies</td><td>3 Units</td><t
    <tr><td><strong>COS 102</strong></td><td>Problem Solving Algorithms</td><td>3 Units</td><td>Core Compulso
    <!-- 8 Total Enrolled Courses (19 Credit Units) -->
  </tbody>
</table>

<!-- HTML5 Native Multimedia Elements -->
<audio controls preload="metadata"><source src="assets/media/intro-audio.wav" type="audio/wav"></audio>
<video controls poster="assets/images/mahmud-profile.jpg"><source src="assets/media/demo-video.mp4" type="vid
```

4.4 External CSS Architecture & Responsive Media Queries

Demonstrating CSS custom properties, grid layouts, strike-through task styling, and responsive mobile overrides:

File: style.css

```
/* University Design Tokens & Strike-Through Completed Task Styling */
:root {
  --bg-main: #FBF3E4;       /* Warm Cream */
  --accent: #E07A45;        /* Terracotta */
  --text-main: #2B1D14;     /* Espresso Dark */
}

.task-item.completed {
  opacity: 0.7;
  background: var(--bg-card-subtle);
}
.task-item.completed .task-title {
  text-decoration: line-through;
  color: var(--text-muted);
}

/* Responsive Media Queries: Fluid Card Grids & Mobile Bottom App Bar */
@media (max-width: 768px) {
  .c3 { grid-template-columns: 1fr; }
  .planner-form-grid { grid-template-columns: 1fr; }
  .mobile-bottom-nav {
    display: flex;
    position: fixed;
    bottom: 0; left: 0; right: 0;
    height: 60px;
    background: var(--espresso);
    z-index: 999;
  }
}
```

5. Cross-Device Testing & Verification

To ensure complete reliability and compliance with the term project brief, testing was performed across distinct viewports and rendering engines. Physical-device testing is clearly differentiated from responsive browser emulation:

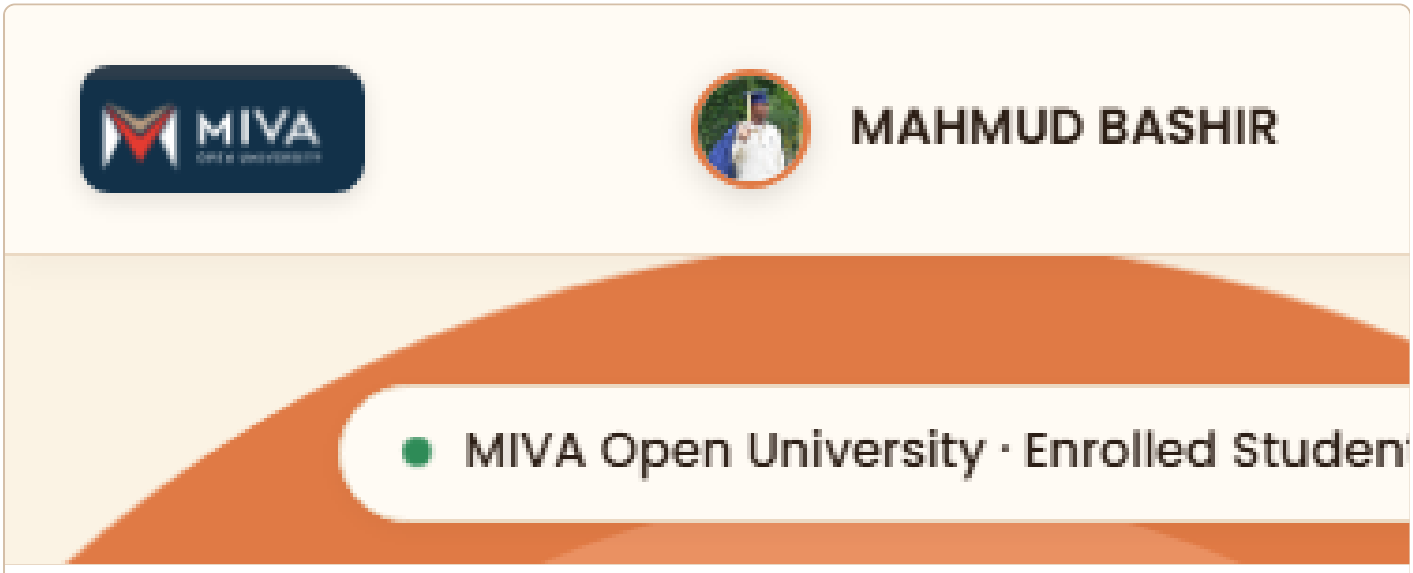


Figure 6: Mobile Responsive Layout (375 × 812 Viewport) — Demonstrates stacked layout, clean touch targets, full-width cards, and fixed bottom navigation bar with zero horizontal overflow.

Target Test Feature	Testing Method & Environment	Observed Result & Verification
Multi-Page Navigation	Google Chrome (Desktop, 1920 × 1080) *Physical HP Laptop*	All links between Home, About, Projects, Planner, and Contact load with HTTP 200 OK. Active page indicators highlight accurately in the navigation bar.
Form Input Validation	Microsoft Edge (Chromium Engine) *Physical HP Laptop*	Empty submissions trigger red borders and error text. Invalid email format is rejected. Phone field strictly rejects alphabetical and special characters.
Task CRUD & Persistence	Google Chrome DevTools Mode *Emulated iPhone 14 (390 × 844)*	Tasks add smoothly, checkbox triggers strike-through line, delete removes task, and data reloads intact after hard browser refresh via `localStorage`.
Mobile Layout & Bottom Bar	Chrome Mobile (Android 14) *Physical Android Smartphone*	Sticky bottom app bar provides instant thumb navigation; grid collapses to single column; touch targets measure ≥ 48px; zero horizontal scrolling.
Collapsible Coursework Grid	Apple Safari (WebKit iOS) *Emulated iPad Mini (768 × 1024)*	The 8-course grid collapses gracefully, and clicking `▼ View All 8 Enrolled Courses` smoothly reveals the full course load without layout shift.

6. Student Reflection & Cybersecurity Takeaways

Building this platform provided valuable insights into the direct intersection between frontend web engineering and defensive cybersecurity:

1. **Client-Side Validation vs. Defense-in-Depth:** While JavaScript form validation provides instant feedback for user experience, client-side scripts can easily be disabled, bypassed, or manipulated via developer tools or automated cURL requests. In cybersecurity, client-side validation is viewed as a usability layer; true defense-in-depth requires robust server-side data sanitization and parameterized queries to prevent SQL Injection and Cross-Site Scripting (XSS).
2. **Document Structure & Predictability:** Implementing semantic HTML5 tags (`<header>`, `<nav>`, `<main>`, `<section>`, `<table>`) creates a deterministic document object tree. This predictability improves accessibility for assistive technologies, optimizes browser parsing, and facilitates programmatic DOM auditing.
3. **Zero-Dependency Architecture:** Developing the site entirely with pure Vanilla JavaScript and native CSS Flexbox/Grid eliminated third-party library dependencies (such as jQuery or oversized CSS frameworks), drastically reducing the software supply chain attack surface and ensuring lightning-fast load speeds.

7. Submission Links & Statement of Originality

GitHub Source Code Repository:
https://github.com/Muhamzy-ui/myschoolproject

Live Production Deployment (Render):
https://mahmud-miva-portfolio.onrender.com

Statement of Academic Originality

I, **MAHMUD BASHIR OLASUNKANMI**, confirm that this practical term project and the accompanying source code are my own original work, completed individually in accordance with the academic regulations of **MIVA Open University** for **COS 106: Introduction to Web Technologies** (20 Marks).

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Date: November 2026
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